

WASP-10b

R-1047

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_151229 · created 2026-07-30T09:21:44+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2457385.6385 +/- 0.0026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.123 +/- 0.013
Transit depth (Rp/Rs)^2	1.5 +/- 0.32 [%]
Orbital Inclination (inc)	88.65 +/- 0.74
Airmass coefficient 1 (a1)	0.981 +/- 0.013
Airmass coefficient 2 (a2)	0.011 +/- 0.0043
Scatter in the residuals of the lightcurve fit is	1.3 %
Best Comparison Star	#2 - [252, 131]
Optimal Aperture	3.99
Optimal Annulus	12.43
Transit Duration (day)	0.0887 +/- 0.0034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.123 ± 0.013 vs 0.1592 (deviation 2.45σ)
Duration fit vs published	0.0887 d vs 0.0946 d (deviation 1.52σ)
Mid-time O–C	-0.5 min
Depth SNR	8.3
Residual scatter	1.3 %

Light curve

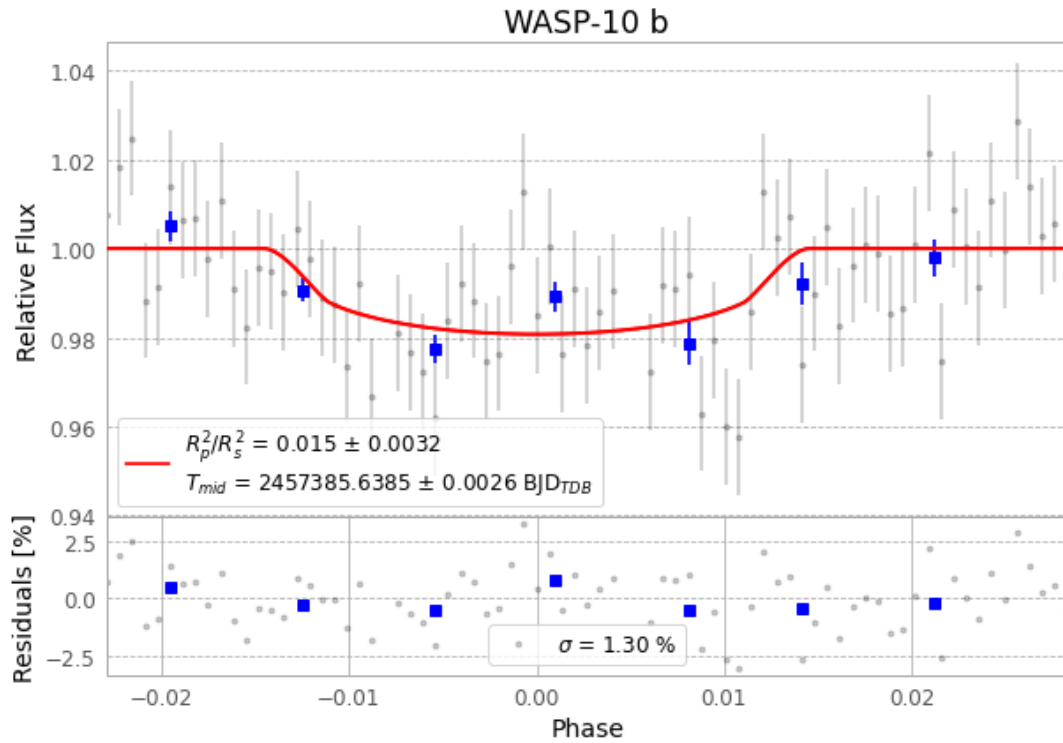


Figure 1 — FinalLightCurve_WASP-10 b_2015-12-28.png (SHA-256 7f46358aa6d69a9d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [256, 230], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 08377f58855f2eb1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), WASP-10151229013612.FITS → WASP-10151229052412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-151229.log (f281c2dea8ce...), FinalLightCurve_WASP-10 b_2015-12-28.png (7f46358aa6d6...), FinalLightCurve_WASP-10 b_2015-12-28.csv (a70ee9e58c4a...)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 09:21Z.